I will do transcript obtaining, text mining, and sentiment analysis of a YouTube video on the Harvard

Salata Institute titled "What Could Go Wrong? Harvard Climate Action Week 2024".

Install or Import Libraries

In [1]:
```
#install libraries
!pip install youtube-transcript-api textblob wordcloud nltk PyPDF2 numpy pandas mat
!pip install beautifulsoup4 requests
```

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```

In [2]:

```python
#Import libraries for use for for transcript extraction, sentiment analysis, data o
from youtube_transcript_api import YouTubeTranscriptApi
from textblob import TextBlob
from wordcloud import WordCloud, STOPWORDS

import pandas as pd
import matplotlib.pyplot as plt
import re
import requests
```

Choose a YouTube Video

In [3]:

```python
#I selected a 2024 climate panelist YouTube video hosted by Harvard's Salata Instit
# url: https://www.youtube.com/watch?v=d1vymVAAfR4
video_id = "d1vymVAAfR4"

print("Video ID:", video_id)
```

```
Video ID: d1vymVAAfR4
```

Extract the transcript from the YouTube Video

In [4]:

```python
def transcript_entry_to_dict(entry):
    """Attempt to convert transcript entry to a dictionary.

    """
    if isinstance(entry, dict):
        return {
            "text": entry.get("text", ""),
            "start": entry.get("start", None),
            "duration": entry.get("duration", None)
        }
```

```
    return {
        "text": getattr(entry, "text", ""),
        "start": getattr(entry, "start", None),
        "duration": getattr(entry, "duration", None)
    }


try:
    ytt_api = YouTubeTranscriptApi()
    transcript = ytt_api.fetch(video_id)
    transcript_data = [transcript_entry_to_dict(entry) for entry in transcript]

    print("Transcript successfully loaded.")
    print("Number of transcript entries:", len(transcript_data))

except Exception as e:
    transcript_data = []
    print("The transcript could not be loaded.")
    print("Possible reasons:")
    print("- The video does not have captions or a transcript.")
    print("- The transcript is disabled.")
    print("- The video ID is incorrect.")
    print("- There is a temporary connection or access issue.")
    print()
    print("Error message:")
    print(e)
```

```
Transcript successfully loaded.
Number of transcript entries: 1502
```

In [5]:
```
#Create a dataframe of the transcript
df = pd.DataFrame(transcript_data)

if df.empty:
    print("No transcript data is available. Please try another video ID.")
else:
    display(df.head())
```

	text	start	duration
0	welcome everyone my name is Dan	0.680	6.280
1	shrag I am a professor here in Earth and	3.359	5.561
2	planetary Sciences across the street in	6.960	3.200
3	the School of Engineering and applied	8.920	3.240
4	sciences and also at the Harvard C	10.160	5.599

Clean the transcript text

In [6]:
```
def clean_text(text):
    """Remove extra spaces and line breaks."""
    text = str(text)
```

```
    text = text.replace("\n", " ")
    text = re.sub(r"\s+", " ", text)
    return text.strip()


if not df.empty:
    df["clean_text"] = df["text"].apply(clean_text)
    display(df[["text", "clean_text"]].head(5))
```

	text	clean_text
0	welcome everyone my name is Dan	welcome everyone my name is Dan
1	shrag I am a professor here in Earth and	shrag I am a professor here in Earth and
2	planetary Sciences across the street in	planetary Sciences across the street in
3	the School of Engineering and applied	the School of Engineering and applied
4	sciences and also at the Harvard C	sciences and also at the Harvard C

Combine the Transcript fragments into One Text

In [7]:
```
if not df.empty:
    full_text = " ".join(df["clean_text"].dropna().tolist())

    print("Total characters in transcript:", len(full_text))
    print()
    print(full_text[:1500])
else:
    full_text = ""
    print("No transcript text is available.")
```

```
Total characters in transcript: 54888

welcome everyone my name is Dan shrag I am a professor here in Earth and planetary S
ciences across the street in the School of Engineering and applied sciences and also
at the Harvard C school and I put this panel together called what could go wrong now
the purpose of this panel I want to explain just a little bit it's not to wallow in
terrifying things and scare everybody to death that's not the purpose although that
may a side effect um I'm a climate scientist who's worked on climate change for a ve
ry long time and and the way I see the climate problem is that we're doing an experi
ment on the planet that hasn't been done for a very long time I my core area was in
paleoclimate and so I studied earth's climate over hundreds thousands millions even
billions of years and so what we're doing is extraordinary higher carbon dioxide lev
els and other greenhouse gases is pushing the earth into a place that hasn't been fo
r millions of years and at a rate that's never happened before the rate of change an
d so I think what a lot of people fail to understand about climate change is that th
ere's a lot we don't understand and that's partly because this has never happened be
fore it's truly extraordinary what is going on the experiment we're doing on the pla
net and so surprises are inevitable and unfortunately some people think that and hav
e in the past used our lack of understanding as an excuse for well maybe it's not go
ing to be so bad unfortunately in my experience almost everything we d
```

Conduct Sentiment and Polarity Analysis Using Textblob

In [8]:
```
def get_polarity(text):
    return TextBlob(text).sentiment.polarity


def get_subjectivity(text):
    return TextBlob(text).sentiment.subjectivity


if not df.empty:
    df["polarity"] = df["clean_text"].apply(get_polarity)
    df["subjectivity"] = df["clean_text"].apply(get_subjectivity)

    display(df[["clean_text", "polarity", "subjectivity"]].head(10))
```

	clean_text	polarity	subjectivity
0	welcome everyone my name is Dan	0.8000	0.9
1	shrag I am a professor here in Earth and	0.0000	0.0
2	planetary Sciences across the street in	0.0000	0.0
3	the School of Engineering and applied	0.0000	0.0
4	sciences and also at the Harvard C	0.0000	0.0
5	school and I put this panel together	0.0000	0.0
6	called what could go	0.0000	0.0
7	wrong now the purpose of this panel I	-0.5000	0.9
8	want to explain just a little bit it's	-0.1875	0.5
9	not to wallow in terrifying things and	-1.0000	1.0

In [9]:
```
# Show results in an easier to interpret format by converting the numeric scores to
def sentiment_label(score):
    if score > 0:
        return "Positive"
    elif score < 0:
        return "Negative"
    else:
        return "Neutral"


if not df.empty:
    df["sentiment"] = df["polarity"].apply(sentiment_label)
    display(df[["clean_text", "polarity", "sentiment"]].head())
```

	clean_text	polarity	sentiment
0	welcome everyone my name is Dan	0.8	Positive
1	shrag I am a professor here in Earth and	0.0	Neutral
2	planetary Sciences across the street in	0.0	Neutral
3	the School of Engineering and applied	0.0	Neutral
4	sciences and also at the Harvard C	0.0	Neutral

Interpret the Sentiment Results of the Beginning Content of the Video

This covers only a small introductory portion of the video's transcript, but I chose this video because it was a scientific

discussion about a heated issue. Except for the welcome introduction, the beginning transcript lines have a neutral

polarity.

On subjectivity, the results are mostly zero as this is a scientific talk. I expect the non-neutral sentiments in the full video

will have positively-sentimented polite praise for the panelists, and the negative sentiment sections will be statements

about troubling environmental forecasts.

More specifically for the first few lines:

Row 0 has a strong positive polarity of .8 and a strong subjectivity level of .9. It's an opening statement that welcomes the audience.

Row 1,2,3,and 4 have no polarity and no sentiment since they are statements of fact.

These results suggest that after a positive statement of greeting, the beginning of the presentation consists of factual statements.

In [ ]:

```
### Sentiment Summary:
```

In [10]:

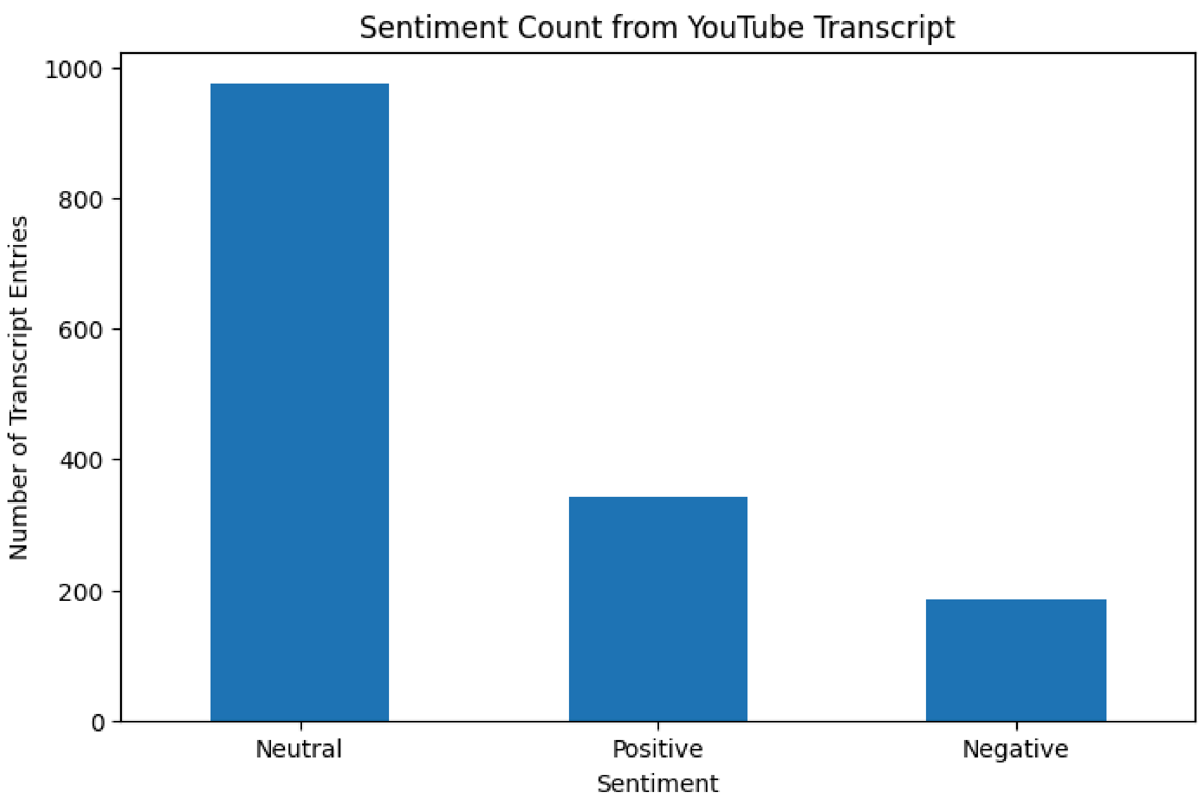
```
if not df.empty:
    sentiment_counts = df["sentiment"].value_counts().reset_index()
    sentiment_counts.columns = ["Sentiment", "Count"]
    display(sentiment_counts)
```

	Sentiment	Count
0	Neutral	975
1	Positive	341
2	Negative	186

Sentiment Visualization:

In [12]:

```
#This displays a histogram of the transcripts sentiments categorized into neutral,
if not df.empty:
    sentiment_counts.plot(kind="bar", x="Sentiment", y="Count", legend=False, figsi
    plt.title("Sentiment Count from YouTube Transcript")
    plt.xlabel("Sentiment")
    plt.ylabel("Number of Transcript Entries")
    plt.xticks(rotation=0)
    plt.show()
```



Interpretation: the video overall is largely neutral due to its scientific nature. There are slightly more

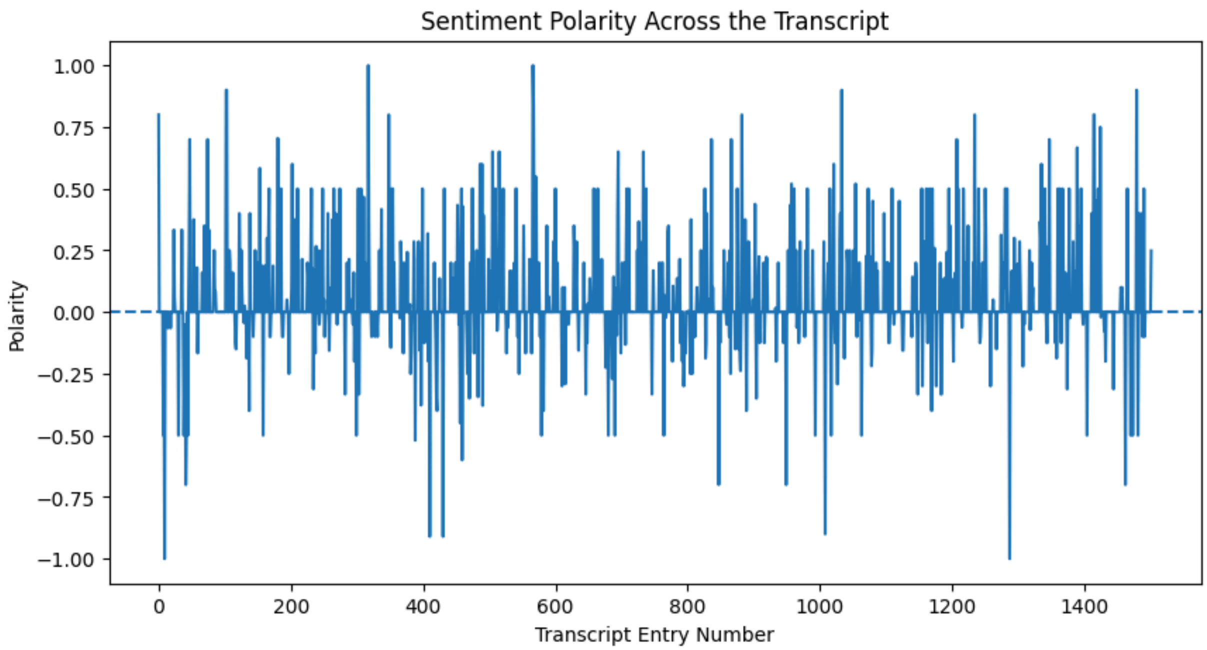
positive sentiments than negative sentiments.

Visualization of Polarity Over Time:

In [13]:

```
if not df.empty:
    plt.figure(figsize=(10, 5))
```

```
    plt.plot(df.index, df["polarity"])
    plt.axhline(0, linestyle="--")
    plt.title("Sentiment Polarity Across the Transcript")
    plt.xlabel("Transcript Entry Number")
    plt.ylabel("Polarity")
    plt.show()
```



There was not a clear trend over time in sentiment. In other words, the transcript did not start out positive and

get gloomier as it went on, for instance.

Generate a Word Cloud:

In [14]:
```
if full_text.strip():
    custom_stopwords = set(STOPWORDS)

    wordcloud = WordCloud(
        width=900,
        height=450,
        background_color="white",
        stopwords=custom_stopwords,
        collocations=False
    ).generate(full_text)

    plt.figure(figsize=(12, 6))
    plt.imshow(wordcloud, interpolation="bilinear")
    plt.axis("off")
    plt.title("WordCloud of YouTube Transcript")
    plt.show()
else:
    print("No transcript text is available for the WordCloud.")
```

Setback: This did not appear to remove words such as "um" and "uh", which should be stopwords. However, it did seem to remove stopwords such as "a", "and", and "the". I got the hilarious result that the most common words in the discussion were "um" and "uh".

Conclusion: The actual content of the video predicts a dire environmental future. However, the scientific and

polite nature of panelists' interactions masks the video's actual grim content. This shows some limitations

of sentiment analysis.

In [ ]:

```

```